

Efficient multi-hop wireless broadcast protocol in vehicular networks using automated threshold function design

Michael Slavik¹, Imad Mahgoub^{1,2,*} and Mohammed Alwakeel¹

¹*Sensor Networks and Cellular System (SNCS) Research Center, University of Tabuk, Saudi Arabia*

²*Department of Computer and Electrical Engineering and Computer Science, Florida Atlantic University, FL, USA*

SUMMARY

Vehicular networking applications often use multi-hop wireless broadcasting as a primary data dissemination mechanism. Therefore, protocols that efficiently and thoroughly propagate application data while adapting to a wide range of network density, vehicle distribution pattern, channel quality, and other conditions are critical for vehicular communications. Here, we design the Statistical Vehicular Broadcast (SVB) protocol to efficiently distribute data via multi-hop broadcast in vehicular networks.

First, we present an automated optimization technique for the design of threshold functions in statistical broadcasting methods. Next, we compare and analyze known statistical techniques, including different fundamental methods, assessment delay algorithms, and failsafe mechanisms. All combinations of these techniques are given threshold functions optimized using the proposed automated procedure then are evaluated in a wide range of simulations. High-level statistical design principles and recommendations are established based on analysis of these results. Finally, we apply those principles to design SVB. It is evaluated in JiST/SWANS and is shown to achieve a high target reachability level while consuming less bandwidth than similar protocols across urban and highway vehicular networking scenarios. Copyright © 2014 John Wiley & Sons, Ltd.

Received 19 November 2013; Revised 31 January 2014; Accepted 26 March 2014

KEY WORDS: wireless broadcast; statistical broadcast; broadcast storm; VANET

1. INTRODUCTION

Vehicular Ad-hoc NETWORK (VANET) applications involving information dissemination are likely to rely on multi-hop wireless broadcast as a key communications method. Implementation of multi-hop broadcast by blindly retransmitting broadcast packets (flooding) is inefficient and can quickly saturate the network, a condition known as a *broadcast storm* [1]. Often relying on multi-hop broadcast ability, VANETs are particularly susceptible to this problem and in need of efficient multi-hop broadcast protocols. These VANET communications supplement cellular communications to support a broad range of vehicular networking applications [2].

Broadcast protocol design is an active area of research. VANETs present several challenges that are not necessarily present in fixed networks or Mobile Ad-hoc NETWORKS (MANETs) in general. The distribution of nodes and wireless channel quality in VANETs can be highly variable because of the relatively high level of mobility. On isolated highways, vehicles are constrained along one-dimensional paths. In regions with more roads in proximity of each other, vehicles may appear more uniformly distributed in two dimensions. Broadcast protocols for VANET must accommodate all these scenarios.

*Correspondence to: Imad Mahgoub, Department of Computer and Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431, USA.

†E-mail: imad@cse.fau.edu

Multi-hop broadcasting algorithms can be classified into at least statistical and topological methods [3]. Topological protocols use the network topology (full knowledge of all network links within two hops) to select nodes to forward broadcast messages. The key advantages of topological protocols are that they reduce broadcast redundancy in a way that is agnostic to geographical node distribution and many can be proven to not degrade reachability compared to flooding, provided that communications are perfectly reliable [4].

However, topological protocols also have disadvantages. High levels of node mobility in VANETs mean that the network topology is highly variable. Overhead messaging used to discover network topology may become a significant load on the network bandwidth if protocols require timely topological information. Statistical broadcast protocols typically do not use this rapidly changing neighborhood information. Statistical protocols measure the value of one or more locally available variables and make a decision to rebroadcast based on a comparison of the measured value to a threshold value. As a result, most statistical protocols are not guaranteed to propagate a message to all nodes in the target network. They are, however, highly resilient to changes in density, distribution pattern, and channel quality, provided that the protocol adjusts the decision threshold value in such a way as to operate efficiently across these environments.

Because of the nature of vehicular networks, we conclude that statistical methods are a promising platform for broadcast protocols in VANET. The challenge then is to design a statistical protocol that exhibits high efficiency while achieving required reachability levels across the broad range of vehicular networking conditions. This work addresses this challenge by proposing a novel optimization scheme for the design of statistical broadcast protocol decision threshold functions, applying the technique to many of the known statistical broadcast methods, and proposing the Statistical Vehicular Broadcast (SVB) protocol based on these results.

The contributions of this work are summarized as follows:

- *Automated Design of Statistical Protocols*: A generalized design methodology using a multi-variable optimization algorithm to discover efficient threshold functions that takes into account all desired network conditions that nodes are able to measure. The framework includes methods for black-box optimization, threshold function representation, and fitness evaluation. The methodology is applied to combinations of statistical broadcast methods and ultimately used to produce the SVB protocol.
- *Comparison and Analysis of Statistical Techniques*: We present a taxonomy of statistical broadcast techniques including different fundamental methods, assessment delay algorithms, and failsafe mechanisms. These techniques are combined and evaluated using extensive simulations. Results are analyzed, and novel statistical protocol design recommendations are given based on the empirical data.
- *SVB Protocol*: Informed by the comparison of the various statistical methods, the automated design procedure is applied to the distance-to-mean broadcast heuristic with distance-weighted random assessment delay to produce the SVB protocol.

The remainder of the paper is organized as follows. In Section 2, we discuss related work. Section 3 surveys and classifies statistical broadcasting techniques. Section 4 describes the proposed statistical broadcast optimization procedure. Section 5 presents results comparing combinations of the various statistical broadcast techniques and outlines the SVB protocol. Section 6 presents simulation results of SVB with other existing protocols, and Section 7 gives concluding remarks.

2. RELATED WORK

The focus of this work is statistical multi-hop broadcasting in VANETs. The fundamental statistical methods, stochastic, counter, distance, and location, are described in [1]. Several papers have addressed the need to make stochastic broadcast adaptive to density [5–7]. Density-adaptive versions of counter are given in [8, 9], and density-adaptive versions of both counter and location methods are given in [10]. We proposed the DADCQ protocol, a statistical protocol based on the distance method that is adaptive to node density, distribution pattern, and channel quality in [11]. An analytical

model of broadcasting in 802.11p multi-channel systems, upon which many of these broadcast protocols may be applied, is given in [12].

Multi-hop broadcast in VANET mainly supports two types of broadcast applications. One type is safety applications such as Collective Collision Avoidance (CCA) where in the event of an emergency an alert message is propagated as quickly as possible into oncoming traffic [13]. Broadcast protocols for these applications are usually designed to propagate messages in one or two directions with high reliability and minimum latency. Examples of these protocols are [14, 15], and [16]. The focus of this work is the second type of VANET broadcast application such as traffic data dissemination. These applications require data to be disseminated to a large area as efficiently as possible, with less stringent reliability and delay requirements than CCA-type applications. Protocols to support these applications are more like traditional MANET broadcast protocols, except they must handle the high node mobility and variable node distribution of VANET environments.

In the VANET data dissemination context, several broadcasting techniques have been proposed. Tonguz *et al.*, using a combination of the counter and stochastic methods and distance-biased assessment delay, proposed the p -persistence method [17, 18]. A more comprehensive study of these methods in highway scenarios is rolled into a scheme called DV-CAST [19]. DV-CAST has been shown to have performance issues in urban environments, so the authors propose a variant called UV-CAST designed for urban environments [20]. p -persistence is used as a comparison protocol in this work.

Bako *et al.* adopt a stochastic broadcast method originally proposed in [21] for VANET [22]. Notable from this work is the Advanced Adaptive Gossiping, a hybrid protocol that uses two-hop topological data to set forwarding probabilities used as in a traditional stochastic broadcast algorithm. Another similar protocol that uses stochastic broadcast in conjunction with two-hop topological data is the Neighbor Coverage based Probabilistic Rebroadcast (NCPR) protocol [23]. NCPR is also used in the evaluations here.

Osafune *et al.* propose MHVB [24] and E-MHVB [25], which use the distance method with constant threshold value to suppress rebroadcasts. Korkmaz *et al.* propose Urban Multi-hop Broadcast (UMB) [26], which, like the CCA broadcast protocols, is fundamentally uni-directional, but does contain provisions for branching at intersections to cover wider areas.

Multi-Point Relaying (MPR) [27] is a well-known broadcast protocol in MANET that uses network topology information to select rebroadcasting nodes using a localized Connected Dominating Set (CDS) estimation heuristic. An extension of MPR that is more accommodating to high mobility called E-MPR is proposed in [28]. Ros *et al.* propose stacking a broadcast acknowledgment protocol on top of a CDS-based protocol to create the ABSM scheme [29].

This work is an extension work on applying machine learning to the design of threshold functions [30]. The previous work has been extended with a taxonomy and comprehensive study of statistical broadcasting techniques. Analysis of those results produces several high-level statistical design principles, which are then used to design a new multi-hop broadcast protocol. Preliminary results from this work are presented in [30].

3. STATISTICAL BROADCAST ALGORITHMS

Many multi-hop broadcasting protocols follow the pattern that (1) nodes self-elect whether or not to retransmit a received broadcast and (2) the decision is made by comparing a locally measured value to a decision threshold. The simplest example is stochastic broadcast. In this method, when nodes receive a broadcast message, they each generate a random value and compare to a threshold p to decide to retransmit it or not. Thus, each node independently decides to rebroadcast or not, and the decision depends on a critical threshold value. We classify methods that follow this pattern as *statistical* broadcast methods.

In this section, we first describe the different statistical methods used in this work. Each method is defined by the local variable that nodes measure and compare to a threshold to make rebroadcasting decisions. Many of the methods require an assessment delay algorithm in the process of measuring the variable, so we next describe several techniques for determining assessment delays. Next, we describe the process of beaconing, which is used by many rebroadcasting protocols to measure local

Table I. Fundamental statistical methods.

Variable	Description	Decision criteria
Stochastic (S)	Uniform random number in $[0, 1]$	$S < Threshold$
Counter (C)	Number of times messages is received	$C < Threshold$
Distance (D)	Distance to nearest transmitting neighbor	$D > Threshold$
Distance-to-mean (M)	Distance to spatial mean of transmitting neighbors	$M > Threshold$

network topology. Finally, we discuss the critical role of the decision threshold in the performance of statistical protocols.

3.1. Statistical methods

In this work, we utilize four different statistical methods: stochastic, counter, distance, and distance-to-mean. These methods are summarized in Table I and described in the following sections.

3.1.1. Stochastic method. Stochastic broadcast, also known as probabilistic broadcast or gossiping, is a simple method that reduces the number of nodes that participate in retransmitting a broadcast message by selecting rebroadcasters randomly among the population of nodes. In our formulation, each node generates a uniform random number S in the interval $[0, 1]$ whenever a new broadcast message is received. If S is less than the decision threshold value, the node will rebroadcast the message. An assessment delay is not required to sample S , although an assessment delay may still be used to affect the order in which nodes retransmit.

3.1.2. Counter method. The counter method is based on the heuristic that if a node has received a broadcast message many times, it will likely not improve the broadcast propagation if it retransmits it again. Thus, the variable being measured in the counter method is the number of times a broadcast message has been received, C . Measuring C requires that nodes perform an assessment delay after receiving each message so see how many more times its neighbors will transmit the message. Because C is an integer, the threshold does not have as fine a degree of control as in the other statistical methods listed here. To add granularity, we layer the stochastic method on top of C for our counter method protocol: if the value of C when the assessment delay expires is less than or equal to the threshold t , the node will retransmit. If $C < t + 1$, then the node will rebroadcast if $S < t - \lfloor t \rfloor$, where S is a sample from a uniform random variable in $[0, 1]$ as in the stochastic method. Otherwise, the node will not rebroadcast. The result is that all nodes that have received the message less than $\lfloor t \rfloor$ times will rebroadcast and some percentage of nodes that received the message exactly $\lfloor t \rfloor$ times will rebroadcast.

3.1.3. Distance method. If a node has received a broadcast message from another node that is located physically near it, then there will be little additional area covered if it chooses to rebroadcast. The distance method heuristic argues from this that nodes that have received a broadcast message from nearby nodes should prefer to not rebroadcast and vice versa. An assessment delay is used to measure the minimum distance from which the message has been received, called D . If D is greater than the threshold value when the assessment delay expires, the node retransmits the message.

3.1.4. Distance-to-mean method. The counter and distance methods are introduced by Ni *et al.* in [1]. In that paper, they also describe the location method, which uses a similar heuristic to the distance method: nodes should favor rebroadcasting if the additional area that will be covered by transmitting is large. With the location method, nodes transmit their geographical location along with messages, and nodes calculate what fraction of their transmit area has not been covered by their neighbors when the assessment delay expires. If this uncovered fraction is greater than a threshold, the node retransmits. Calculating the uncovered fraction is somewhat complex as it involves the union of circle-circle intersections (assuming ideal omnidirectional transmissions). The distance-to-mean method uses the same heuristic but is simpler to calculate [31]. Here, each node uses an

assessment delay to find which neighbors have broadcast the message and stores their positions relative to itself. When the assessment delay expires, it calculates the spatial mean of the positions and measures the distance from its position to the spatial mean, M . If M is above the threshold value, the node retransmits.

3.2. Assessment delay

Most statistical techniques use a delay procedure to measure the method variable. This procedure is referred here as an assessment delay. All the assessment delay methods used in this work follow the same pattern:

- (1) When a node receives a message for the first time, calculate a delay time T and set a backoff timer for that time.
- (2) If the message is received before the timer expires, process the message, calculate a new delay time and restart the timer.
- (3) When the timer expires with no new messages received, use the statistical method to make a rebroadcast decision.

There are many ways to determine a delay time T , and the choice of method can have significant impact on the performance of the broadcast protocol. In all methods used here, there is a maximum backoff delay, T_{max} , and the delay is chosen in the interval $[0, T_{max}]$.

In this work, we experiment with five assessment delay methods:

- *Uniform*: Delay is chosen uniformly randomly from $[0, T_{max}]$.
- *Counter-Biased*: Delay is proportional to the number of times the message has been received.
- *Distance-Biased*: Delay is inversely proportional to the distance to the nearest neighbor from which the message has been received.
- *Density-Biased*: Delay is proportional to the number of neighbors the node has.
- *Reverse Density-Biased*: Delay is inversely proportional to the number of neighbors the node has.

Many broadcast protocols use a uniform assessment delay, which is also referred to as a Random Assessment Delay (RAD) [32]. The distance-biased assessment delay is also common and used notably in p -persistence [18], the broadcast method employed by DV-CAST [19] and UV-CAST [20]. Section 5 provides a comprehensive comparison of these techniques.

3.2.1. Failsafe mechanism. Related to the assessment delay, some protocols such as p -persistence [18] and DCB [4] also use a fallback timer that we refer to as the failsafe mechanism. At the end of the assessment delay, if a node decides not to retransmit, it will set a failsafe timer for T_{max} . If this timer expires and the message has not been received again (suggesting it is likely none of its neighbors elected to retransmit), it will retransmit the message itself. This affords some degree of reliability in case a node chose not to rebroadcast, but may also decrease protocol efficiency. In the evaluations in Section 5, we show the affect of using the failsafe mechanism on protocol performance.

3.3. Beaconing

Next, we will describe the mechanism for communicating topological information between nodes, called beaconing. Nodes broadcast a short message every T_b seconds called a beacon. Unlike the multi-hop broadcast messages that become propagated to the entire network, beacons are broadcast from a node to its neighbors only and are never retransmitted. Nodes send beacons at a constant fixed rate, so in any interval of T_b seconds, every node in the network will send exactly one beacon and receive exactly one beacon from each of its neighbors (assuming no packet loss). The timing of beacons is in no way synchronized between nodes, except that all nodes use the same value of T_b .

The size and type of data contained in beacons depends on the broadcast protocol. If nodes need no topological information, as in p -persistence, beaconing can be disabled entirely. If nodes need only to count the number of neighbors, then beacons can be sent with no additional data. If nodes

require geographical information, then nodes insert their current location in the beacon. If two-hop neighborhood information is required, as in topological protocols, nodes insert their ID and current list of neighbors in the beacon. Note that it is not only the statistical method that determines what type of information is needed but also the assessment delay method and threshold function input variables (Section 4.2).

If the network is static, the beacon interval T_b can be very long and nodes will still have reasonably accurate topological data. However, when nodes are mobile, as in VANET, the beacon interval needs to be reduced so that beacons received near the beginning of an interval are not stale by the end of the interval. All the statistical protocols designed here use T_b of 30 s.

3.4. Decision threshold and reachability

Statistical broadcast protocol performance is determined in large part by the decision threshold. The threshold value is re-calculated dynamically at each node for broadcast message reception using the most current topological information available to the node evaluated by a fixed threshold function. This value is compared to the method variable to decide whether or not that node will retransmit the broadcast message. If the threshold is set too aggressively, the message will not propagate through the network. If it is set too conservatively, too many nodes will elect to rebroadcast and the protocol will not be efficient.

Analysis in [33] shows a rough approximation of this phenomenon for the case of the distance-to-mean method (Figure 1). Reachability here is defined as the fraction of nodes in the network that receive the broadcast message. This result is a simplification, as many parameters that may affect the practical behavior of a wireless multi-hop broadcast system are necessarily neglected; however, the graph highlights the essence of the threshold design problem for statistical broadcast methods in the form of a phase transition. In the supercritical phase, reachability is nearly one in all simulation runs. Around some critical value of the rebroadcast threshold, the reachability quickly transitions from 1 to 0 and the variance of the reachability spikes, indicating the reachability is highly variable from run to run. In the subcritical phase, message propagation is suppressed so that only nodes near the source will receive the message.

As these results show, the performance of statistical methods can vary dramatically based on the threshold value or function. Therefore, we focus a considerable amount of attention on the design process for obtaining efficient threshold functions (Section 4).

3.5. Usage of statistical broadcast protocols in VANET

Statistical broadcasting protocols built on the framework described in this work are designed to minimize bandwidth consumption subject to a constraint on broadcast reachability. Reachability is specified to surpass some high average level, but further guarantees on message delivery are not provided. Message propagation delay is not explicitly optimized.

Applications requiring these performance characteristics include the broad class of general information dissemination, such as traffic flows, public advisories, weather, and commercial adver-

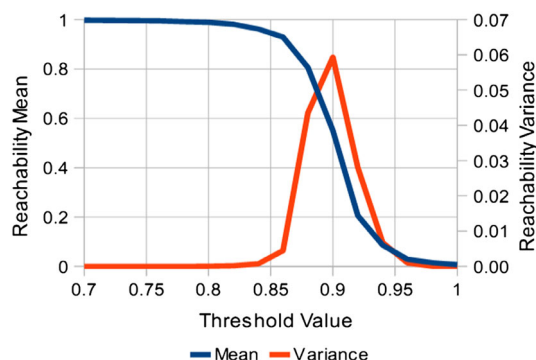


Figure 1. Reachability versus threshold value in distance-to-mean method [33].

tisement. Here, data being sent are non-critical and may include a large amount data sent from many data gatherers. Thus, reducing bandwidth consumption is preferred to absolute reliability or minimal propagation delay. Statistical broadcasting methods as described in this work are not recommended for applications such as cooperative collision avoidance, where reliability must be guaranteed and transmission delay must meet strict requirements, without providing additional mechanisms or design strategies to meet those requirements.

All the protocols described in this work can be used in a wider network layer routing scheme to provide broadcast message routing for application types given previously. While some timing details may need to change to accommodate virtual channelization, these protocols are compatible with IEEE 802.11p, so should be practical to deploy in existing VANET systems.

4. DESIGN OF STATISTICAL BROADCAST PROTOCOLS

The previous section gave an overview of statistical protocols and showed the importance of the threshold value on the performance of the protocol. In this section, we outline our proposed threshold function design procedure. Because of the impact of threshold value on performance, the threshold design procedure is a key design process for statistical protocols.

At its core, the threshold design procedure is an optimization problem. The goal is to find a threshold function that maximizes protocol performance. Implementation of the optimization framework requires a few steps. First, a parameterized functional form is needed to represent the threshold function, in this case, a multi-layer perceptron. Next, select a set of locally measurable variables as inputs to the threshold function. Finally, design a method for evaluating a threshold function instance using a battery of simulation tests that ultimately produce a single fitness value. With these pieces in place, a standard multi-variable global optimization algorithm can be applied to search for threshold functions with maximum fitness. The following sections describe these steps in detail.

4.1. Threshold function representation

The optimization algorithms used in this procedure work by searching a multi-dimensional space for the point that has the highest fitness. Our role in this process is to provide the optimization algorithm with a fitness value for any point it wishes to evaluate. Thus, we first need a way to translate an array of numbers representing the point into a threshold function.

There are many potential ways to make this translation. The multi-layer perceptron (MLP) is convenient for this application because it is capable of approximating any function to arbitrary accuracy (given enough hidden layer perceptrons), yet is simple to implement and calculate. Other functional forms, however, may be used to suit the preference of the designer. The formula for the MLP used here is

$$y(x) = f \left(\sum_{j=1}^n v_j f \left(\sum_{i=1}^m w_{j,i} x_i + w_{j,m+1} \right) + v_{n+1} \right) \quad (1)$$

We will have three input variables, so $m = 3$. The activation function f is a sigmoid $f(x) = 1/(1 + e^{-x})$. The weight vectors, w and v , are the parameters of the functional form. In total, there are $(m + 1)n + (n + 1) = 5n + 1$ weights. Because the number of weights is equivalent to the number of dimensions in the search space, large n (number of hidden neurons) may make it difficult for the optimization algorithm to find good solutions. Through experimentation varying n from 1 to 10, we find that $n = 2$ provides a good balance between function flexibility and search difficulty.

To translate from a vector given to us by the optimization algorithm to an MLP instance, we simply map the vector elements directly onto w and v .

4.2. Threshold function inputs

The optimal value of the decision threshold is dependent on many external factors. Ideally, any factors impacting the optimal value of the threshold should be reflected in the set of threshold function input variables, so that the protocol can be adaptive to them. We find that the following variables

provide sufficient information to the threshold function for the resulting protocol to be adaptive across a broad range of network conditions: number of neighbors N , the quadrat statistic Q , and the Rician fading factor K . N is measured easily using beacon messages. Q and K are detailed in the succeeding discussion.

4.2.1. Quadrat statistic Q . The quadrat method attempts to measure how evenly spaced a set of points are. A point distribution can either be regularly spaced, such as in a lattice, positioned in clumps, or somewhere in between. The quadrat method addresses the problem by examining the frequency pattern of the point distribution. The quadrat statistic, Q , is defined as the ratio of the variance of the frequency distribution to the mean. Here, nodes record the locations of neighbors relative to themselves and use these points to calculate a value of Q . This value is used as an input to the threshold function to provide information about the spacing of nodes in the system.

The value of Q is a positive real number with a central value of 1. When the node distribution is two-dimensional uniformly random, the value of Q is near 1. If $Q < 1$, then the variance of the frequencies is less than the mean, indicating the nodes are more evenly spaced than is expected in uniformly distributed networks. If the nodes are clustered together, then the variance will be larger than the mean and $Q > 1$. Please see [11] for further details on the quadrat method and its use in statistical broadcasting.

4.2.2. Rician fading factor K . To model fading in VANET, we use Rician fading. In Rician fading, the received signal is assumed to have a strong component from line-of-sight among scattered multipath signals. A parameter, K , quantifies the relative strength of the line-of-sight signal. When $K = 0$, also called Rayleigh fading, there is no strong line-of-sight signal. When $K = \infty$, fading has no effect so all nodes within transmission range of each other can reliably communicate.

For purposes of this work, we assume nodes have the means to estimate K .

4.3. Evaluating a threshold function

The optimization algorithm requires a method for evaluating the performance of a candidate threshold function to a single value called fitness. Multi-hop wireless broadcast is a complex system, so simulation is used to assess the performance of a given threshold function. WiBDAT, a fast high-level multi-hop broadcast simulation tool, was developed specifically for this purpose [34]. WiBDAT uses stochastic models to represent fading and collisions so that only the network layer broadcast protocol is simulated directly. This makes the simulator fast and scalable, which is important when designing threshold functions because many simulation runs are required. The tradeoff for execution time is simulation detail and accuracy. However, we have found that this tradeoff is necessary to complete a threshold function design in a practical amount of time. In the evaluation and analysis section, we will use JiST/SWANS [35], a standard detailed wireless network simulator, to perform detailed protocol evaluations. WiBDAT is used only for this design process.

For each candidate threshold function generated by the optimization algorithm, we run a set of test cases meant to exercise the function in a broad range of network conditions. In all, we use 10 scenarios outlined in Table II. The maximum transit radius in all scenarios is 10. Each scenario tests a range of node densities, for a total of 32 simulations. Each simulation uses three iterations, resulting in a total of 96 runs for each threshold function evaluation. With WiBDAT, the total evaluation time is about 6 s on a quad-core Intel i7 machine with 6 GB of RAM.

Table II. Battery of tests to evaluate candidate threshold functions.

Distribution pattern	K	Range of N
Rectangle, 60×60	0, 10, 20	10–30
Line, 60×1	0, 10, 20	20–50
Thin rectangle, 60×5	0, 20	20–50
Grid	0, 20	15 to 30

When the evaluation is complete, we then measure reachability and transmit density and calculate an overall fitness. For reachability, we use the Pareto model described in [33] to calculate the probability that the threshold function will achieve a reachability level greater than 95%, called L . We set our application-required value for L also at 95% (meaning the application requires at least 95% of nodes receive broadcasts in 95% of trials). The transmit density, λ_{tx} , is defined as the number of transmitting neighbors averaged across all nodes that receive the broadcast message. Values of λ_{tx} range from zero to a maximum value of the node density. The values of L and λ_{tx} are averaged across all simulation runs, then an overall fitness level is calculated as follows.

$$Fitness = 1000 \frac{\min\{0.95, L\}}{0.95} - 10\lambda_{tx} \quad (2)$$

The two fitness weights, 1000 and 10, are chosen by experimentation to reflect the magnitude of the relative importance of the two factors. The functional purpose of a multi-hop broadcast protocol is to distribute information through the network, so reachability confidence is given a weight two orders of magnitude higher than transmit density. Thus, a small degradation in message propagation has a large impact on the fitness of a threshold function, as desired. The value of L is adjusted so that equal reward is given for any threshold function to exceed the confidence level target of 95%. If a threshold function meets the reachability target, it receives 1000 from that part of the equation. Second, we reward lower bandwidth consumption by subtracting the transmit density with a relatively small weight.

4.4. Optimization algorithm

Finally, we employ a standard multi-variable optimization procedure to search the threshold function space for functions with the highest fitness value. We experimented with several optimization packages and found that most algorithms we tested produced high-quality solutions. Thus, for this work, we choose one algorithm, Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [36], provided by the Python PyBrain package (version 0.3) [37] which we use with its default parameters. For further information on the details of this CMA-ES implementation, please refer to the documentation for PyBrain.

The PyBrain optimization framework searches automatically through the function space for the point with maximum fitness. At each function evaluation, we take the point vector, translate it to an MLP, execute the WiBDAT test battery, and return the calculated fitness value. When the CMA-ES algorithm is completed, a vector with the best discovered fitness is returned. This vector is translated into an MLP and used to define the statistical protocol being optimized.

5. EVALUATION OF STATISTICAL BROADCAST METHODS

In this section, we present our comparative simulation analysis of statistical protocols generated using the automated design process of Section 4. For completeness, we generate threshold functions for all combinations of the techniques listed in Section 3. We use four different statistical methods: stochastic, counter, distance, and distance-to-mean; five assessment delay methods: uniform, counter-biased, distance-biased, density-biased, and reverse density-biased; and two failsafe modes: enabled and disabled. All combinations are used, so the total number of independent broadcast methods evaluated in this section is 40.

We first discuss the simulation environment and then report the results. The results are organized by statistical mechanism, assessment delay method, and failsafe mechanism. Finally, the top protocol is selected and used to create the SVB protocol.

5.1. Setup

The simulation environment uses two main tools. We first use VanetMobiSim [38] to generate realistic vehicular mobility traces. These traces are inputted into the distribution of JiST/SWANS [35] produced by Ulm University to give an accurate network simulation.

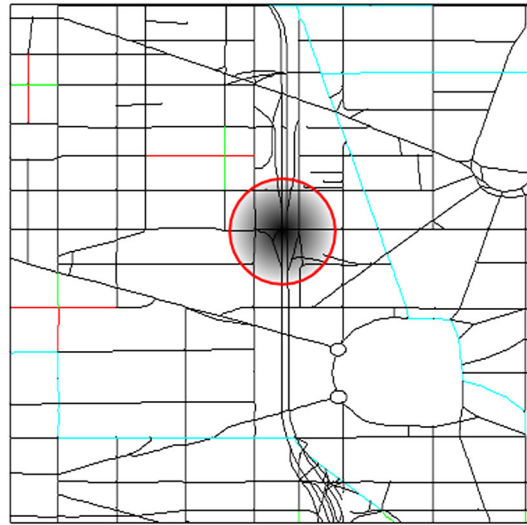


Figure 2. Urban scenario map with overlaid maximum transmission range.

We evaluate the protocols under two different mobility scenarios. The first is an urban scenario based on a 2 km square region of Washington, D.C., shown in Figure 2, built using TIGER data published by the U.S. Census Bureau. The second is a highway scenario that restricts node movements to a 1800 m square. Node movement uses Intelligent Driver Model with Lane Changing with a given maximum speed. These two mobility scenarios serve to test a wide range of node distribution patterns.

Each network simulation begins with a 60 s warm up period followed by 1000 s of broadcast activity. During the active period, a new 128-byte broadcast message is originated about every 1 s by a randomly chosen node. Signal propagation is modeled with two-ray ground propagation with Rician fading. JiST/SWANS supports 802.11a PHY and MAC models, and the Ulm University improvements for VANET support the 5.9 GHz band for 802.11p support. Thus, the PHY and MAC simulation is fully compliant with IEEE 802.11p [39]. Radio parameters are configured 5.89 GHz, 3 Mbps, with about 400 m transmission range. JiST/SWANS models both packet collisions and channel fading. Nodes use 802.11 MAC layer signaling (removing association and authentication, as in 802.11p) and IPv4 layer 3 addressing. The maximum RAD time used in the protocols is 50 ms, consistent with similar proposals such as p -persistence [17]. Following analysis given in [40], the beacon interval is set to 30 s.

5.2. Comparison results

The protocols are tested in three phases. The first phase varies the number of nodes from 250 to 500 in increments of 50, keeping the Rician fading parameter K set to 10 and the maximum speed set to 25 m/s. The second phase varies the maximum speed from 15 to 40 m/s in increments of 5 m/s with nodes fixed at 250 and $K = 10$. Finally, K is varied from 0 to 20 in increments of 5 with nodes fixed at 250 and maximum speed set fixed at 25 m/s. Each of these phases is run on both the urban map and the highway map, so each protocol is submitted to a total of 34 different simulation scenarios.

We use three different metrics to show the results. The first, reachability, shows the average fraction of nodes that receive a broadcast message. The second shows the number of rebroadcasts (not including beacons) per node that received the broadcast. For example, a value of 0.5 indicates that there were 0.5 retransmissions on average for each node that received the source message. This metric considers how many nodes were selected to rebroadcast and ignores overhead due to beacons. The third metric is the number of bytes sent (including beacons) per node that received the message.

Tables III and IV show the top performing protocols with respect to reachability and rebroadcasts per covered node. Figure 3 shows the performance of the different statistical broadcast techniques

Table III. Top 10 statistical method combinations by reachability.

Statistical method	Assessment delay method	Failsafe	Reachability	Rebroadcasts per covered node	Bytes sent per covered node
Counter	Distance-biased	On	0.977	0.254	58.3
Counter	Counter-biased	On	0.976	0.495	111.1
Counter	Density-biased	On	0.976	0.295	67.2
Distance	Distance-biased	On	0.975	0.182	42.6
Stochastic	Distance-biased	On	0.975	0.553	123.9
Counter	Uniform	On	0.975	0.284	65
Distance-to-mean	Distance-biased	On	0.975	0.216	50
Distance-to-mean	Counter-biased	On	0.974	0.342	77.7
Stochastic	Uniform	On	0.973	0.571	127.8
Stochastic	Density-biased	On	0.973	0.815	181.4

Table IV. Top 10 statistical method combinations by rebroadcasts per covered node.

Statistical method	Assessment delay method	Failsafe	Reachability	Rebroadcasts per covered node	Bytes sent per covered node
Counter	Distance-biased	Off	0.915	0.156	37.1
Distance	Distance-biased	On	0.975	0.182	42.6
Distance	Uniform	On	0.957	0.191	44.6
Distance-to-mean	Distance-biased	Off	0.971	0.194	45.1
Distance-to-mean	Uniform	On	0.959	0.194	45.1
Distance	Density-biased	Off	0.968	0.195	45.6
Counter	Uniform	Off	0.955	0.195	45.5
Distance	Distance-biased	Off	0.968	0.203	47.2
Distance	Uniform	Off	0.967	0.204	47.4
Distance-to-mean	Uniform	Off	0.97	0.205	47.7

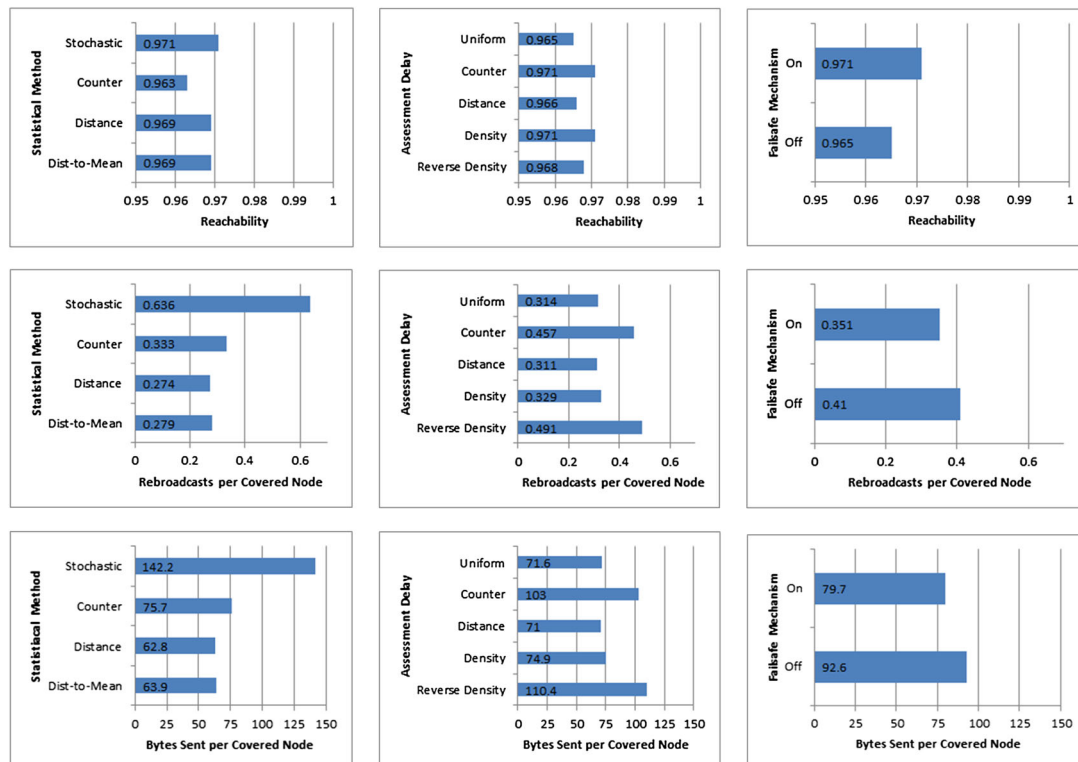


Figure 3. Comparison of statistical protocol performance, grouped by statistical method, assessment delay method, and failsafe mode.

averaged across all the protocols that use them. For example, the upper-left plot in Figure 3 shows the reachability of the four statistical methods, averaged across all 10 combinations of assessment delay and failsafe mode. Presented this way, the graphs allow us to quickly compare the average overall performance of the various techniques with respect to the different metrics.

The reachability results show that nearly all the protocols exhibit penetration ratios around our design target of 95% nodes covered in 95% of broadcasts. Among the top 10 combinations in Table III, all four statistical methods and four of five assessment delay methods are represented. All 10 imply the failsafe mechanism, providing evidence for the expected result that the failsafe mechanism improves protocol reachability. In all, these results show that the proposed design methodology almost always produces a protocol capable of meeting the target level of reachability.

Rebroadcasts per covered node and bytes sent per covered node are both measures of the efficiency of the broadcast protocols, with lower scores implying higher efficiency and less bandwidth consumption. These results show some clear trends. Stochastic broadcast is the least efficient method and is not represented in the top 10 combinations for rebroadcast per covered node (Table IV). The most efficient combination is the counter method with distance-biased assessment delay and no failsafe, but the reachability of that combination is below the target level so it is discarded. Among the rest, distance and distance-to-mean are about equal and show the best performance, both averaged across all assessment delays and failsafe modes and in the top performing combinations.

An interesting pattern emerges related to the amount of information used by the statistical method. The method using no external information, stochastic, performs worst. Distance and distance-to-mean, requiring received distance estimation and node location that are aggregated across multiple message receptions, perform the best. In between, the counter method uses only a count of how many messages are received. It appears that, all else being equal, the statistical methods that measure the external environment more closely perform better. Note that in our formulation here, node location is used as part of the threshold function evaluation, so using less external measurements in the statistical method does not provide a significant advantage.

Among assessment delay methods, distance-biased appears to be the best performing, followed closely by uniform. Density-biased is also a close third on average but is only used in one of the top 10 combinations for bandwidth consumption. The counter-biased and reverse density-biased assessment delay methods are counterproductive on average and are not used in the top 10 protocols for bandwidth consumption.

The distance method, distance-to-mean method, and the distance-biased assessment delay are based on the same heuristic: broadcast propagation should favor retransmissions by nodes that most increase the area being covered in the retransmission. Nine of the top 10 protocol combinations in terms of bandwidth consumption use one of these methods as part of the protocol, arguing strongly that among the heuristic arguments for various statistical broadcasting techniques discussed in Section 3.1, it is the most powerful.

The results give a strong endorsement for the failsafe mechanism. Protocols can simultaneously achieve higher reachability and lower bandwidth consumption with the failsafe enabled. Note that this does not imply that adding the failsafe mechanism to an existing protocol will lower its bandwidth consumption. Most likely, the reachability will increase but so will the bandwidth consumption. However, if the protocol is optimized using the procedure of Section 4, the optimization algorithm discovers that it is possible to set the threshold value more aggressively with the failsafe mechanism enabled but still achieve the target reachability level.

Overall, we reach the following conclusions regarding the performance of statistical techniques evaluated here:

- (1) Any statistical broadcast protocol based on the techniques described here can achieve a high level of reachability, given an appropriate threshold function.
- (2) Protocols that employ a mechanism designed to maximize additional area covered by retransmitters by making rebroadcasts by nodes that are geographically near each other less likely are the best at minimizing bandwidth consumption.
- (3) The retransmit failsafe mechanism, when included in the threshold function optimization process, can both increase protocol reachability and reduce bandwidth consumption.

5.3. Statistical vehicular broadcast protocol

The main goal of this comparative study is to inform the choice of the best statistical techniques for our proposed protocol. Because all the designed protocols met the specified reachability target, the best performing protocols can be chosen as the one with the lowest bandwidth consumption (as those are our two primary design goals). Tables III and IV show the top individual combinations of statistical methods.

Considering both reachability and bandwidth consumption, we select the distance method with distance-biased assessment delay and failsafe mechanism as the combination of methods for the proposed SVB protocol. The SVB rebroadcast algorithm is shown in Algorithm 1.

Algorithm 1 SVB

Definitions:

T_{max} : Maximum assessment delay time

d : Distance to neighbor from which message received

r : Maximum expected transmit radius

N : Current number of neighbors

Q : Current quadrat statistic

K : Estimated Rician fading parameter

```

1: if Broadcast message received for the first time then
2:   Initialize  $D = 1$ 
3:
4:   {Assessment Delay Algorithm}
5:   repeat
6:     Calculate distance to sender  $d$ 
7:     Update  $D = \min\{D, d/r\}$ 
8:     Set a backoff timer with duration  $T_{max}(1 - D)$ 
9:   until Message received again
10:
11:   {Statistical Method}
12:   Calculate  $t = y(N, Q, K)$  according to Equation 1 and the MLP instance shown in Table V
13:   if  $D > t$  then
14:     Rebroadcast the message
15:   else
16:
17:     {Failsafe Mechanism}
18:     Set a backoff timer for  $T_{max}$ 
19:     if Backoff timer expires without the message having been received again then
20:       Rebroadcast the message
21:     end if
22:   end if
23: end if

```

Table V. Solution multi-layer perceptron instance used in Statistical Vehicular Broadcast protocol.

Perceptron	Weights
Hidden 1	0.324, 0.356, -0.286, -1.261
Hidden 2	-0.407, 1.285, 0.187, 0.493
Output	0.797, 1.159, 0.548

6. COMPARISON OF SVB WITH EXISTING PROTOCOLS

In this section, we compare the performance of the proposed SVB protocol with existing protocols. First, we describe the comparison protocols, then present the simulation results.

6.1. Comparison protocols

We compare SVB to three existing protocols: p -persistence, NCPR, and DADCQ. p -persistence is a well-known broadcast protocol proposed for VANET. ETSI is proposed using a broadcast protocol based on p -persistence in its ITS communications system [41]. NCPR is a state-of-the-art hybrid statistical and topological protocol that has recently been proposed for use in mobile ad-hoc networks. DADCQ is a statistical protocol based on the distance method we proposed in [11]. DADCQ is similar in structure to SVB, except its threshold function was designed using a manual procedure. Thus, improvements in SVB over DADCQ represent the gains made with the automated optimization procedure proposed in Section 4. Brief summaries of p -persistence and NCPR are given here. For more details of DADCQ, please see [11].

6.1.1. p -persistence. The first protocol we will use for comparison is p -persistence, originally proposed by Tonguz *et al.* [17]. p -persistence is interesting because it utilizes three different statistical methods simultaneously. Foremost, p -persistence uses the counter method with a threshold value of one. When a node receives a message, it first sets a backoff timer and waits to see if the message is received again. If it is, the retransmission is canceled. If not, the node rebroadcasts.

Calculation of the backoff period involves two other statistical methods. First, nodes decide if they are high priority rebroadcasters according to a fixed probability p . In the evaluations that follow, we follow Wisitpongphan *et al.* [18] and use $p = 0.5$. Nodes that do not elect themselves as high priority rebroadcasters set a backoff time of $T = N_s \tau$, where N_s and τ are protocol parameters. This timer serves as a failsafe mechanism.

Finally, nodes that elect themselves to have high priority use distance-biased assessment delay. Specifically, the backoff time is calculated as $T = \lceil N_s(1 - d/r) \rceil \tau$, where d is the distance to the node from whom the message was received and r is the transmission radius.

Thus, in our nomenclature, p -persistence is a hybrid counter and stochastic method that uses a distance-biased assessment delay and a failsafe mechanism. However, unlike the statistical protocols described here, threshold values in p -persistence are fixed.

For the evaluations in this section, p -persistence protocol parameters are set to $p = 0.5$, $N_s = 5$, and $\tau = 5$ ms.

6.1.2. NCPR. The Neighbor Coverage based Probabilistic Rebroadcast protocol is a stochastic broadcast protocol that uses two-hop neighborhood information to set the threshold value [23]. The protocol uses the concept of *additional coverage ratio* (R_a), defined as the fraction of a node's neighbors that have (or are expected to have) received the broadcast message. Because each node has two-hop neighborhood information, when node u receives a message from node s , it knows which of its neighbors should also have received the message from node s because it knows the neighbors of s . In this manner, each node tracks R_a for each broadcast message and uses this value in determining whether to rebroadcast.

Like many broadcast protocols, NCPR uses an assessment delay method. When a node receives a broadcast message for the first time, it calculates its initial value of R_a and sets a backoff timer for $T_{max} R_a$. Unlike the assessment delay method described here, the backoff timer in NCPR is not restarted if the message is received again. When the backoff timer expires, the node makes a rebroadcast decision. For NCPR, $T_{max} = 10$ ms.

Each time the node receives the message, it updates its value of R_a for that message. When the assessment delay expires, the node calculates the threshold value p as $p = 5.1774(\log n) R_a / N$ where N is the number of neighbors and n is the number of nodes in the network. The authors derive this formula from results of Xue *et al.* investigating connectivity in wireless networks [42]. The node then rebroadcasts the message with probability p .

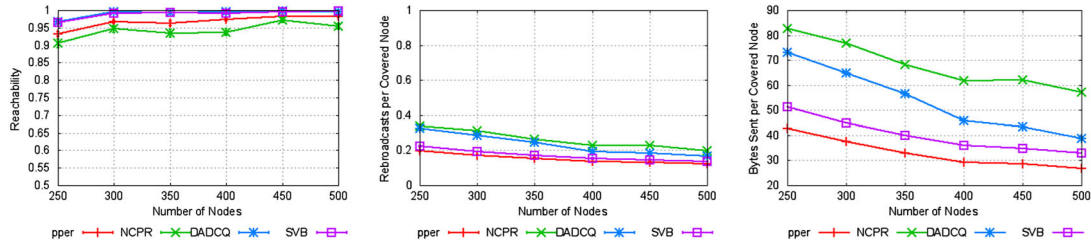


Figure 4. Performance versus number of nodes of the broadcast protocols in the urban scenario.

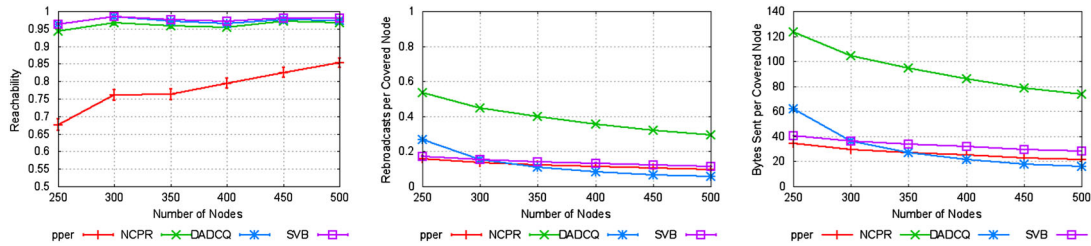
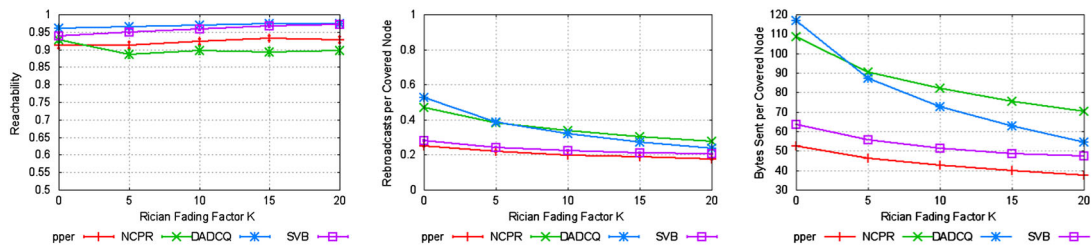


Figure 5. Performance versus number of nodes of the broadcast protocols in the highway scenario.

Figure 6. Performance versus K of the broadcast protocols in the urban scenario.

6.2. Results

The four protocols, p -persistence, NCPR, DADCQ, and SVB, are evaluated using the same simulation setup described in Section 5. Each protocol is submitted to two sets of tests. In the first, the number of nodes is varied from 250 to 500 in increments of 50, holding K at 10. The second varies K from 0 to 20 in increments of 5 with number of nodes fixed at 250. These tests are executed on both the urban and highway maps with maximum vehicle speed set to 25 m/s.

6.2.1. Varying number of nodes. Figures 4 and 5 show the effect of varying the number of nodes on the performance of the protocols. In the urban scenario, all four protocols achieve high reachability, with DADCQ and SVB showing a small advantage. In the highway scenario, however, p -persistence has significantly lower reachability than the other three, indicating it has not been optimized for linearly-oriented networks.

There are two measures of bandwidth consumption: rebroadcasts per covered node and bytes sent per covered node. The former gauges the efficiency with which the broadcast protocol selects retransmit nodes (lower value indicates fewer nodes retransmit), while the latter takes all aspects of the protocol behavior into account, including beacons. p -persistence and SVB consistently select the fewest nodes to retransmit. However, p -persistence consumes less bandwidth overall because it does not require beacons, as shown in the bytes sent per covered node graphs. That it is able to operate efficiently without the use of beacons is one of the strongest advantages of p -persistence.

6.2.2. Varying fading intensity. Figures 6 and 7 show the results of varying the fading intensity from high fading (small values of K) to minimal fading (large values of K). The performance patterns

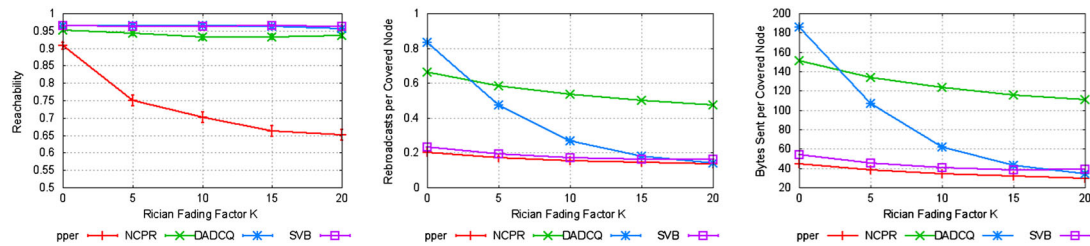


Figure 7. Performance versus K of the broadcast protocols in the highway scenario.

Table VI. Performance results averaged across all scenarios.

Protocol	Reachability	Rebroadcasts per covered node	Bytes sent per covered node
ppper	0.842	0.162	35.1
NCPR	0.937	0.401	95.6
DADCQ	0.977	0.267	61.3
SVB	0.975	0.182	42.6

NCPR, Neighbor Coverage based Probabilistic Rebroadcast; SVB, Statistical Vehicular Broadcast.

shown in the number of neighbors results are borne out here as well. The protocols all exhibit high reachability, with the exception of p -persistence in the highway scenario. SVB and p -persistence are the most efficient, with p -persistence having an advantage in physical bytes sent. NCPR is inefficient in many cases compared to the other three protocols.

With respect to fading, all four protocols have decreased bandwidth consumption as fading decreases (K increases). DADCQ, with a threshold function that heavily weighs the value of K , experiences dramatic performance degradation at high levels of fading. The automated threshold function design procedure described in this work and used to create SVB weighs K more effectively so that the performance of SVB is relatively flat across the range of K .

6.2.3. Summary. Table VI presents the results of the evaluations averaged across all simulations. These overall numbers highlight the trends found in the more detailed results. SVB performs consistently well in all categories. p -persistence has weakness in reachability, and NCPR has weakness in bandwidth consumption.

Algorithmically, DADCQ is very similar to SVB, with the most significant difference being that the DADCQ threshold function was designed manually while the SVB threshold function was discovered empirically using a black-box optimization algorithm. Thus, the performance gains of SVB over DADCQ highlight the advantage of the design procedure proposed in Section 4. While reachability of the two protocols is similar, SVB has a roughly 30% improvement in bandwidth consumption.

7. CONCLUSION

In this work, we have proposed a novel statistical broadcast threshold design procedure employing an automated optimization algorithm with a simulation test battery. We present our taxonomy of statistical techniques, including fundamental statistical methods, assessment delay algorithms, and the failsafe mechanism. All combinations of these techniques are run through the proposed optimization algorithm and evaluated using an extensive JiST/SWANS simulation environment. The results are presented and used to compare the relative performance of the different techniques.

The results of the comparative study are used to design the SVB protocol. SVB uses the distance method with a distance-biased assessment delay algorithm and a rebroadcast failsafe mechanism. The threshold function of SVB is designed using the automated optimization procedure and is adaptive to node density, node distribution pattern, and channel quality.

Simulation results compare the performance of SVB to two existing protocols, p -persistence and NCPR, as well as to a predecessor of SVB called DADCQ. SVB consistently shows high levels of reachability and low bandwidth consumption. Comparison protocols either do not meet the same level of reachability as SVB or do so while consuming more bandwidth. Compared to DADCQ, SVB shows a roughly 30% performance gain in terms of bandwidth consumption, highlighting the benefit of the proposed automated threshold function design procedure.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the support for this work from the Sensor Networks and Cellular Systems Research Center (SNCS) at the University of Tabuk under the grant from the Ministry of Higher Education in Saudi Arabia.

REFERENCES

1. Ni SY, Tseng YC, Chen YS, Sheu JP. The broadcast storm problem in a mobile ad hoc network. *MobiCom '99: Proceedings of the 5th Annual ACM/IEEE International Conference on Mobile Computing and Networking*, New York, NY, USA, 1999; 151–162.
2. Alwakeel MM, Aalo VA. New cell residence time models for mobile leo satellite cellular networks. *IEEE Transactions on Vehicular Technology* 2007; **56**(5):3060–3072. DOI: 10.1109/TVT.2007.900382.
3. Slavik MJ, Mahgoub I. Statistical broadcast protocol design for unreliable channels in wireless ad-hoc networks. *Proceedings of IEEE Globecom 2010 - Wireless Networking Symposium (GC10 - WN)*, Miami, Florida, USA, 2010; 1–5.
4. Lou W, Wu J. Toward broadcast reliability in mobile ad hoc networks with double coverage. *IEEE Transactions on Mobile Computing* 2007; **6**(2):148–163. DOI: 10.1109/TMC.2007.31.
5. Cartigny J, Simplot D, Carle J. Stochastic flooding broadcast protocols in mobile wireless networks, LIFL Univ. Lille 1, 2002. (Available from: <http://citeseer.ist.psu.edu/525199.html>) [Accessed on 23 May 2012].
6. Li XY, Moaveninejad K, Frieder O. Regional gossip routing for wireless ad hoc networks. *Mobile Networks & Applications* 2005; **10**(1-2):61–77. DOI: 10.1023/B:MONE.0000048546.90440.0d.
7. Bako B, Schoch E, Kargl F, Weber M. Optimized position based gossiping in vanets. *IEEE 68th Vehicular Technology Conference (VTC 2008-Fall)*, Calgary, Canada, 2008; 1–5. DOI:10.1109/VETECF.2008.449.
8. al Humoud SO, Mackenzie LM, Abdulai J. Neighbourhood-aware counter-based broadcast scheme for wireless ad hoc networks. *IEEE GLOBECOM Workshops*, New Orleans, LA, USA, 2008; 1–6. DOI:10.1109/GLOCOMW.2008.ECP.45.
9. Mohammed A, Ould-Khaoua M, Mackenzie LM, Abdulai JD. Dynamic probabilistic counter-based broadcasting in mobile ad hoc networks. *2nd International Conference on Adaptive Science Technology (ICAST)*, Accra, GHANA, 2009; 120–127. DOI:10.1109/ICASTECH.2009.5409735.
10. Tseng YC, Ni SY, Shih EY. Adaptive approaches to relieving broadcast storms in a wireless multihop mobile ad hoc network. *IEEE Transactions on Computers* 2003; **52**(5):545–557. DOI:10.1109/TC.2003.1197122.
11. Slavik M, Mahgoub I. Spatial distribution and channel quality adaptive protocol for multihop wireless broadcast routing in vanet. *IEEE Transactions on Mobile Computing* 2013; **12**(4):722–734. DOI:10.1109/TMC.2012.42.
12. Campolo C, Molinaro A, Vinel A, Zhang Y. Modeling prioritized broadcasting in multichannel vehicular networks. *IEEE Transactions on Vehicular Technology* 2012; **61**(2):687–701. DOI:10.1109/TVT.2011.2181440.
13. Huang CM, Lin SY. An early collision warning algorithm for vehicles based on v2v communication. *International Journal of Communication Systems* 2012; **25**(6):779–795.
14. Biswas S, Tatchikou R, Dion F. Vehicle-to-vehicle wireless communication protocols for enhancing highway traffic safety. *IEEE Communications Magazine* 2006; **44**(1):74–82. DOI:10.1109/MCOM.2006.1580935.
15. Ye F, Yim R, Guo J, Zhang J, Roy S. Prioritized broadcast contention control in vanet. *IEEE International Conference on Communications (ICC)*, Cape Town, South Africa, 2010; 1–5. DOI:10.1109/ICC.2010.5501792.
16. Bi Y, Cai LX, Shen X, Zhao H. A cross layer broadcast protocol for multihop emergency message dissemination in inter-vehicle communication. *IEEE International Conference on Communications (ICC)*, Cape Town, South Africa, 2010; 1–5. DOI: 10.1109/ICC.2010.5501865.
17. Tonguz OK, Wisitpongphan N, Parikh JS, Bai F, Mudalige P, Sadekar VK. On the broadcast storm problem in ad hoc wireless networks. *3rd International Conference on Broadband Communications, Networks and Systems*, 2006; 1–11. DOI:10.1109/BROADNETS.2006.4374403.
18. Wisitpongphan N, Tonguz OK, Parikh JS, Mudalige P, Bai F, Sadekar V. Broadcast storm mitigation techniques in vehicular ad hoc networks. *IEEE Wireless Communications* 2007; **14**(6):84–94. DOI: 10.1109/MWC.2007.4407231.
19. Tonguz OK, Wisitpongphan N, Bai F. Dv-cast: a distributed vehicular broadcast protocol for vehicular ad hoc networks. *IEEE Wireless Communications* 2010; **17**(2):47–57. DOI:10.1109/MWC.2010.5450660.
20. Viriyasitavat W, Bai F, Tonguz OK. Uv-cast: an urban vehicular broadcast protocol. *IEEE Vehicular Networking Conference (VNC)*, Jersey City, NJ, USA, 2010; 25–32. DOI:10.1109/VNC.2010.5698266.

21. Kyasanur P, Choudhury R, Gupta I. Smart gossip: an adaptive gossip-based broadcasting service for sensor networks. *IEEE International Conference on Mobile Adhoc and Sensor Systems Conference* 2006; 91–100. DOI: 10.1109/MOBHOC.2006.278671.
22. Bako B, Kargl F, Schoch E, Weber M. Advanced adaptive gossiping using 2-hop neighborhood information. *Global Telecommunications Conference (IEEE GLOBECOM)*, New Orleans, LA, USA, 2008; 1–6. DOI:10.1109/GLOCOM.2008.ECP.1045.
23. Zhang X, Wang E, Xia J, Sung D. A neighbor coverage based probabilistic rebroadcast for reducing routing overhead in mobile ad hoc networks. *IEEE Transactions on Mobile Computing* 2012; **11**(1). DOI: 10.1109/TMC.2011.277.
24. Osafune T, Lin L, Lenardi M. Multi-Hop Vehicular Broadcast (MHVB). *6th International Conference on Its Telecommunications Proceedings*, Chengdu, China, 2006; 757–760. DOI:10.1109/ITST.2006.289011.
25. Mariyasagayam MN, Osafune T, Lenardi M. Enhanced multi-hop vehicular broadcast (MHVB) for active safety applications. *7th International Conference on ITS Telecommunications (ITST '07)*, Sophia Antipolis, France, 2007; 1–6. DOI:10.1109/ITST.2007.4295866.
26. Korkmaz G, Ekici E, Ozguner F. Black-burst-based multihop broadcast protocols for vehicular networks. *IEEE Transactions on Vehicular Technology* 2007; **56**(5):3159–3167. DOI:10.1109/TVT.2007.900493.
27. Qayyum A, Viennot L, Laouiti A. Multipoint relaying for flooding broadcast messages in mobile wireless networks. *Hawaii International Conference on System Sciences* 2002; **9**:3866–3875. DOI: 10.1109/HICSS.2002.994521.
28. Wu C, Kumekawa K, Kato T. A novel multi-hop broadcast protocol for vehicular safety applications. *Journal of Information Processing* 2010; **18**:110–124.
29. Ros FJ, Ruiz PM, Stojmenovic I. Acknowledgment-based broadcast protocol for reliable and efficient data dissemination in vehicular ad hoc networks. *IEEE Transactions on Mobile Computing* 2012; **11**(1):33–46. DOI: 10.1109/TMC.2011.253.
30. Slavik MJ, Mahgoub I. Applying machine learning to the design of multi-hop broadcast protocols for vanet. *The International Wireless Communications and Mobile Computing Conference (IWCMC-2011)*, Istanbul, Turkey, 2011; 1742–1747.
31. Slavik M, Mahgoub I, Sibai FN. The distance-to-mean broadcast method for vehicular wireless communication systems. *International Conference on Innovations in Information Technology (IIT)*, Abu Dhabi, UAE, 2011; 371–374.
32. Williams B, Camp T. Comparison of broadcasting techniques for mobile ad hoc networks. *Proceedings of the ACM International Symposium on Mobile Ad Hoc Networking and Computing (MOBIHOC)*, Lausanne, Switzerland, 2002; 194–205.
33. Slavik MJ, Mahgoub I. Designing statistical multi-hop wireless broadcast protocols using confidence levels from stochastic models of reachability. *The International Wireless Communications and Mobile Computing Conference (IWCMC-2011)*, Istanbul, Turkey, 2011; 1045–1050.
34. Slavik MJ, Mahgoub I, Rathod M. Statistical broadcast protocol design with wibdat: wireless broadcast design and analysis tool. *Proceedings of IEEE Wireless Communications and Networking Conference (WCNC 2010)*, Cancun, Mexico, 2011; 1236–1241.
35. Barr R. Swans – scalable wireless ad hoc network simulator user guide, 2004. (Available from: <http://jist.ece.cornell.edu/>) [Accessed on 3 May 2014].
36. Hansen N, Ostermeier A. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation* 2001; **9**:159–195.
37. Schaul T, Bayer J, Wierstra D, Sun Y, Felder M, Sehnke F, Rückstieß T, Schmidhuber J. PyBrain. *Journal of Machine Learning Research* 2010; **11**:743–746.
38. Fiore M, Harri J, Filali F, Bonnet C. Vehicular mobility simulation for vanets. *40th Annual Simulation Symposium, 2007. ANSS '07*, Norfolk, VA, USA, 2007; 301–309. DOI:10.1109/ANSS.2007.44.
39. WG802.11. IEEE standard for information technology – local and metropolitan area networks – specific requirements – part 11: wireless lan medium access control (mac) and physical layer (phy) specifications amendment 6: Wireless access in vehicular environments. *IEEE 802.11p-2010* 2010.
40. Slavik M, Mahgoub I, Alwakeel MM. Analysis of beaconing message rate in vanet multi-hop broadcast protocols. *9th International Conference on High Capacity Optical Networks and Enabling Technologies (HONET)*, Istanbul, Turkey, 2012; 037–041.
41. WG3 I. Intelligent transport systems (its); vehicular communications; geonetworking; part 4: geographical addressing and forwarding for point-to-point and point-to-multipoint communications; sub-part 1: Media-independent functionality. *ETSI EN 302 636-4-1* 2013.
42. Xue F, Kumar PR. The number of neighbors needed for connectivity of wireless networks. *Wireless Networks* 2004; **10**(2):169–181. DOI:10.1023/B:WINE.0000013081.09837.c0.